BAYESIAN INFERENCE & DECISION THEORY • GROUP PROJECT

A Hierarchical Bayesian Model of Hospital Charges

Partial Pooling Across 201 Facilities in the SPARCS Hospital Inpatient Discharge Dataset

PROJECT TEAM

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The Bayesian Approach

What Makes It "Bayesian"

$$P(\theta \mid \text{data}) \propto P(\text{data} \mid \theta) \cdot P(\theta)$$

Posterior ∝ Likelihood × Prior

Rather than producing a single point estimate, Bayesian inference combines what we believed before seeing the data (the prior) with what the data itself shows (the likelihood) to produce a full posterior distribution over every parameter — capturing uncertainty directly, not as an afterthought.

This is what makes hierarchical structure so natural in a Bayesian framework: facility effects are themselves given a prior (a hyperprior), letting the model learn how much pooling is appropriate directly from the data.

Assumptions of Our Model

Log-normality

log(charges) is approximately Gaussian, as confirmed in EDA

Linearity

Fixed effects combine additively on the log-charge scale

Facility exchangeability

Facility intercepts are draws from a common Normal(0, $\sigma_{facility}$) — no facility is treated as special a priori

Homogeneous residual variance

A single σ is assumed across all observations, regardless of facility or severity

Conditional independence

Given the fixed effects and facility effect, observations are independent

How the Hierarchical Model Works

"Hierarchical" means the model has two connected levels — information flows down from the population to each facility, and each facility's estimate flows back up to inform the population.

Population

1 global model

Sets the overall baseline log-charge and how much facilities are allowed to vary ($\sigma_{facility}$)

→

Facilities

201 facility effects

Each facility gets its own intercept u_j, drawn from the shared population distribution

→

Patients

94,289 encounters

Each patient's predicted charge combines their facility's effect with their own characteristics

Patient-level model

$$\log(\text{charges}_i) \sim N(\mu_i, \sigma)$$
$$\mu_i = X_i\beta + u_{j[i]}$$

Facility-level model

$$u_j \sim N(0, \sigma_{facility})$$

The result is partial pooling: a facility with few patients has an uncertain u_j, so it gets pulled toward the population mean; a facility with thousands of patients has enough evidence to mostly stand on its own.

BACKGROUND

Why Hospital Charges, and Why This Data

Hospital charges for clinically similar patients can vary enormously from one facility to another — driven not just by diagnosis and severity, but by institutional pricing policy, geography, and case mix. Understanding what drives that variation matters to patients, insurers, and policymakers alike.

Our goal: build a statistical model of hospital charges that accounts for both patient-level factors (age, admission type, severity, length of stay) and facility-level pricing variation — while giving small, data-sparse facilities a fair, stable estimate rather than an unreliable one.

THE DATASET

SPARCS

Statewide Planning and Research Cooperative System — New York hospital inpatient discharge records

2.1M	raw patient encounters
205	hospital facilities identified
94,289	encounters in final model dataset
201	facilities retained after cleaning

FRAMEWORK

Why a Bayesian Hierarchical Model

Patients are nested within facilities — and facility sample sizes are wildly unequal. That structure rules out two simpler approaches:

Complete Pooling

One model, ignores facility identity entirely.

Misses real institutional pricing differences.

No Pooling

A separate model fit per facility.

Tiny facilities get wild, unreliable estimates.

Partial Pooling

Each facility gets its own effect, shrunk toward the overall average in proportion to how little data it has.

Small facilities borrow strength; large ones speak for themselves.

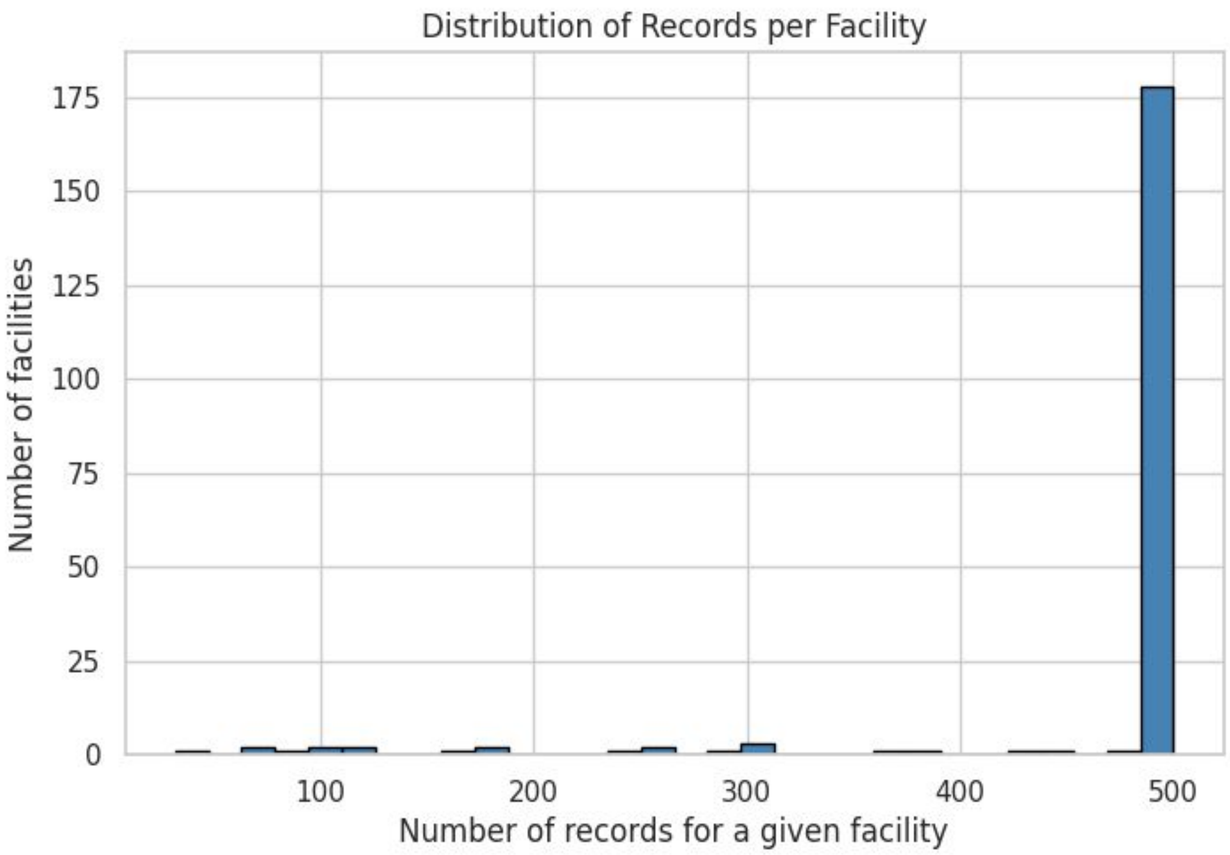
The Bayesian framework lets us build this shrinkage directly into the model: facility effects get a prior distribution (a hyperprior) whose spread is itself estimated from the data — plus full posterior uncertainty on every estimate, not just a point value.

PIPELINE

Data & Preprocessing

1	**Load**	2.1M rows, 33 columns ⟶ 8 relevant columns kept (memory-light load)
2	**Facility ID cleanup**	Drop redacted / missing facility IDs ⟶ 2,090,946 rows, 205 facilities
3	**Group-size handling**	Drop facilities <30 patients; cap large facilities at 500 (random sample) ⟶ 201 facilities
4	**Define outcome**	Total Charges selected — strongest between-facility variation (institutional pricing)
5	**Clean Length of Stay**	Top-coded "120+" values recoded to exactly 120 (right-censoring)
6	**Finalize**	94,289 rows × 8 columns — 7 predictors + charges, ready for modeling

Grouping Structure: Facilities

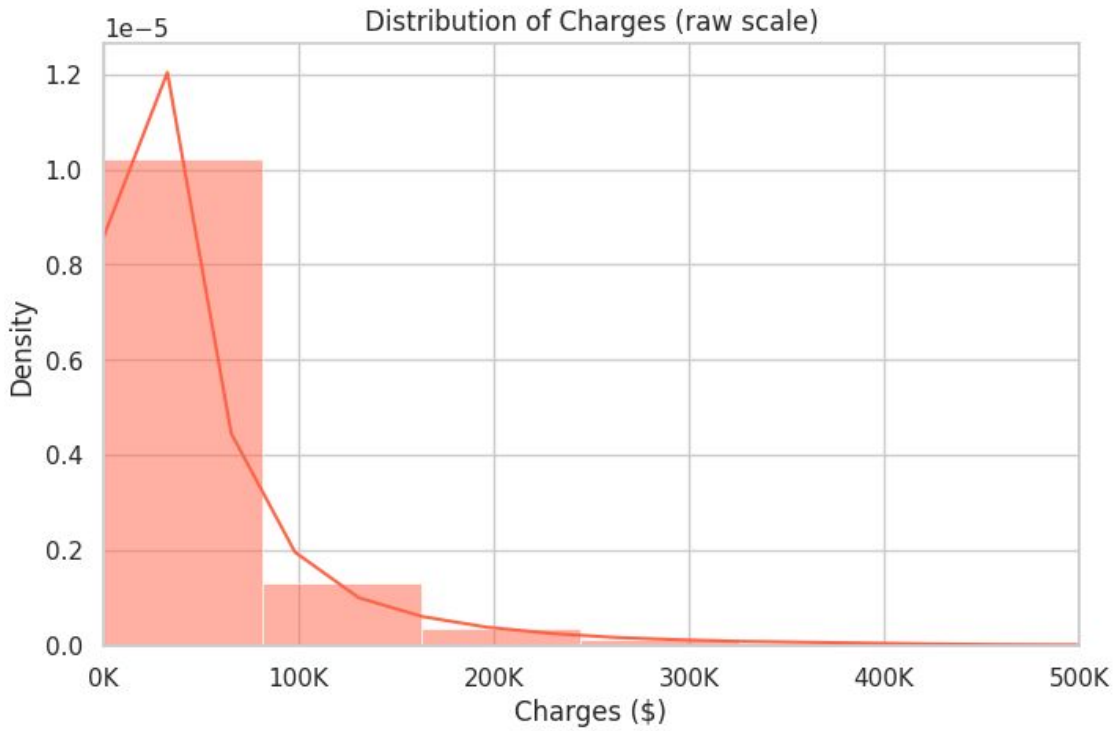


201

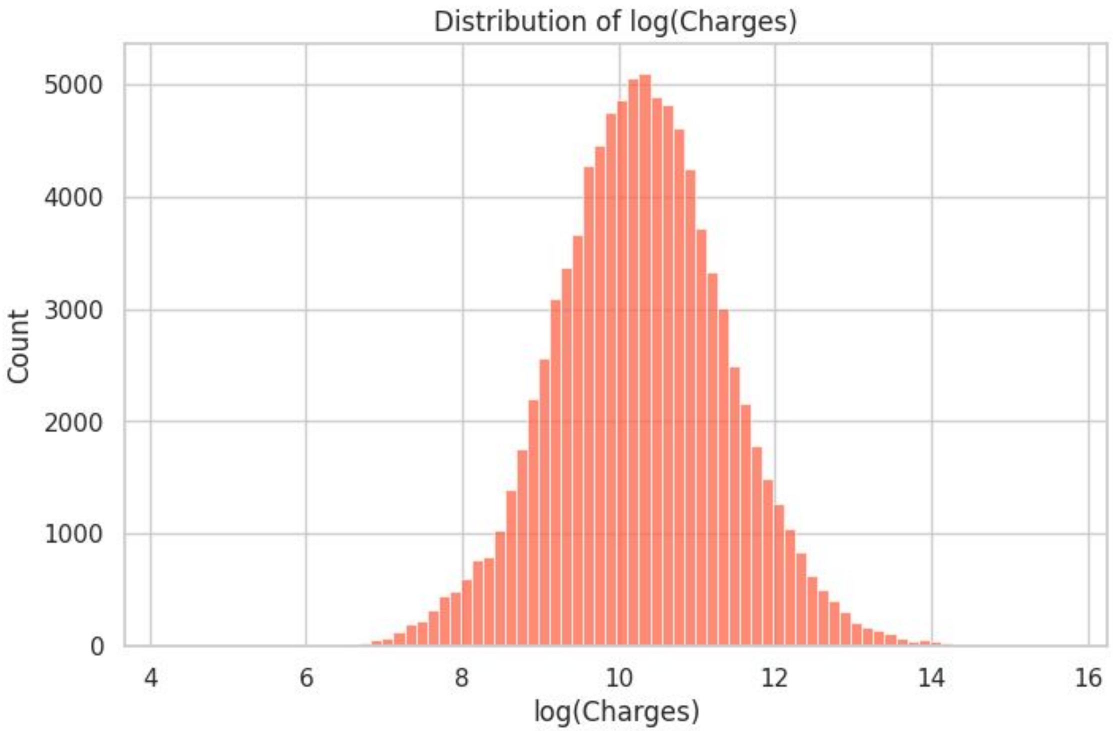
distinct facilities in the final dataset

- Most facilities capped near 500 records (the sampling cap)
- A handful have far fewer — as low as 32 patients
- Records-per-facility: mean 469, min 32, max 500
- Low-count facilities are exactly where hierarchical partial pooling matters most — they benefit most from borrowing strength

Outcome Variable: Charges



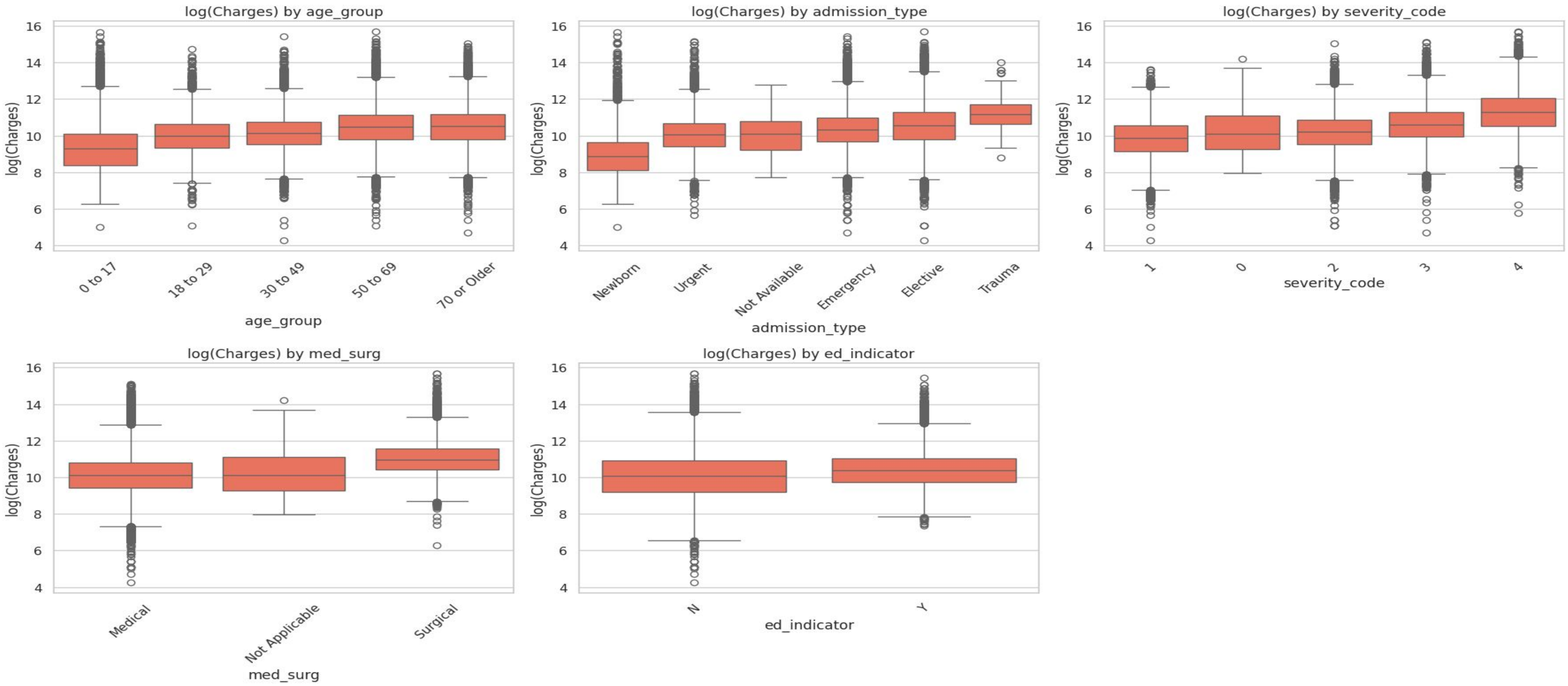
Raw scale — heavy right skew



Log scale — near-symmetric, bell-shaped

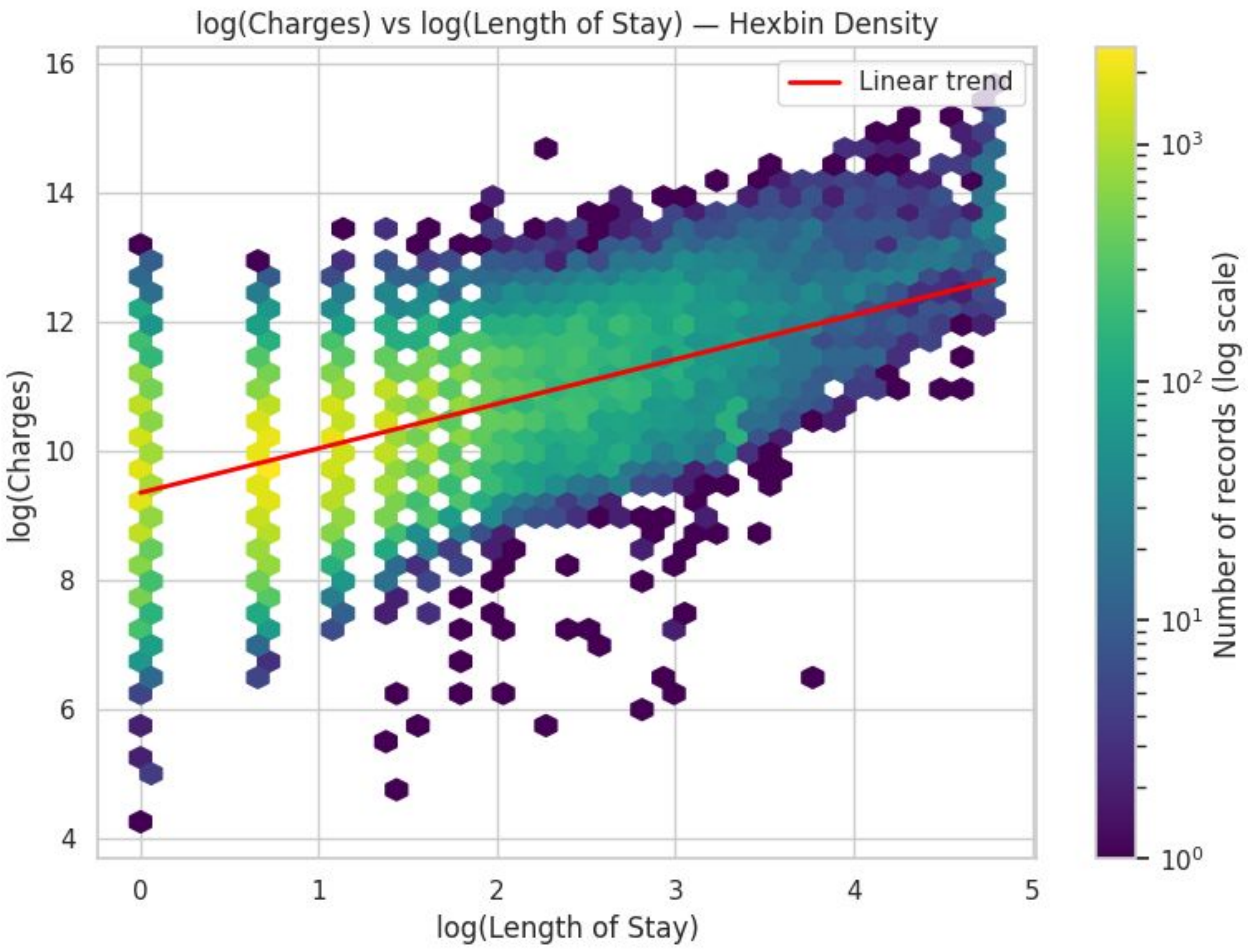
Skewness = 15.22 (raw scale) • Median ≈ \$29,000, Mean ≈ \$56,600, Max ≈ \$6.5M → *Log transform supports a Gaussian likelihood on log(charges)*

Charges vs. Categorical Predictors



All five categorical predictors show visible differences in median log(charges) — age, admission type, severity, and medical/surgical status all carry real signal for the model.

Charges vs. Length of Stay



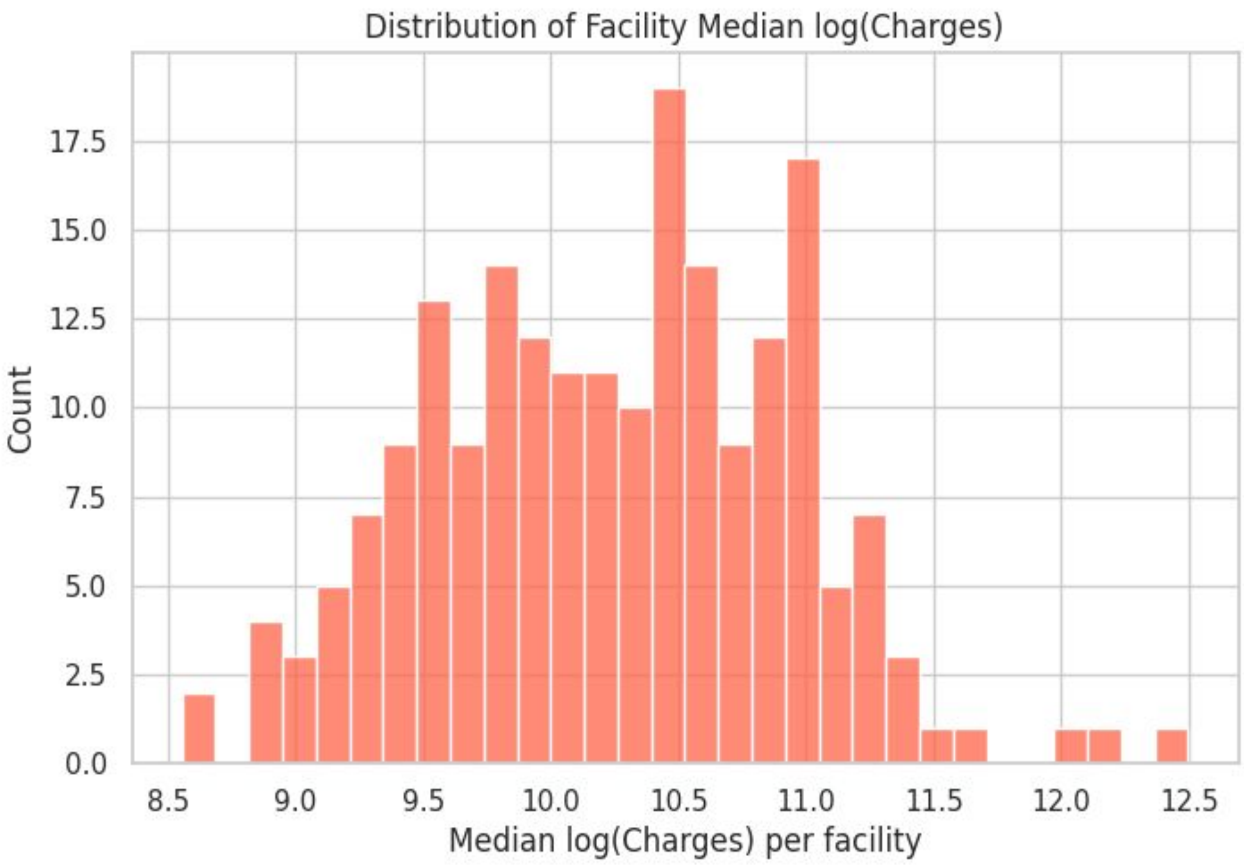
A clear positive relationship

Cost accumulates with each additional day in hospital, as expected. The log-log hexbin view avoids the overplotting and "striping" that a raw scatter would show, since length of stay is discrete.

A note on interpretation

Including length_of_stay as a predictor will likely shrink the estimated facility effect somewhat — LOS mechanically drives charges, so some facility-attributable variation gets absorbed by this term instead.

Facility-Level Variation in Charges



The case for facility as a random effect

If facilities are our grouping variable for the hierarchical model, we need real variation across them for partial pooling to add value. Facility median log(charges) spans roughly 8.5 to 12.5 — clear, substantial spread.

This confirms it

Facility-level medians are spread, not tightly clustered — supporting the inclusion of facility-level random intercepts in the model.

Model Specification

Likelihood

$y_i = \log(\text{charges}_i)$ $\quad\quad$ $y_i \sim \text{Normal}(\mu_i, \sigma)$

Charges are log-transformed to tame the strong right-skew (skewness = 15.22) confirmed in EDA — after transforming, the outcome is approximately Normal.

Linear Predictor

$$\mu_i = \beta_0 + \beta_1 \cdot \text{age\_group}_i + \beta_2 \cdot \text{admission\_type}_i + \beta_3 \cdot \text{severity\_code}_i$$
$$+ \beta_4 \cdot \text{med\_surg}_i + \beta_5 \cdot \text{ed\_indicator}_i + \beta_6 \cdot \text{length\_of\_stay}_i + u_{j[i]}$$

Facility Random Intercept

$u_j \sim \text{Normal}(0, \sigma_{\text{facility}})$ $\quad\quad$ for $j = 1, \ldots, 201$ facilities

Prior Distributions

Weakly informative, data-scaled priors (Bambi's automatic defaults) — regularizing without dominating the likelihood.

Parameter	Prior	Interpretation
Intercept	`Normal(μ=10.28, σ=7.75)`	Centered at ~$29,000; wide prior allows a broad range of baseline charges
age_group	`Normal(μ=0, σ=5.9–9.6)`	Weakly informative; allows moderate age effects
admission_type	`Normal(μ=0, σ=5.8–64.2)`	Very wide for rare categories (Trauma, Not Available) to avoid over-shrinkage
severity_code	`Normal(μ=0, σ=5.7–10.0)`	Allows a meaningful severity gradient
med_surg / ed_indicator	`Normal(μ=0, σ=7.2 / 5.7)`	Reasonable spread for binary-ish category effects
length_of_stay	`Normal(μ=0, σ=0.27)`	Small σ — calibrated to LOS's day-scale units
facility σ (group-level)	`HalfNormal(σ=7.75)`	Allows substantial between-facility variation, as EDA confirmed
σ (residual)	`HalfStudentT(ν=4, σ=1.11)`	Heavy-tailed — robust to outliers in the residual

METHODOLOGY

Estimation Method

The hierarchical model has no closed-form posterior — regression coefficients, facility random effects, and variance parameters combine into an intractable joint distribution. Posterior estimation uses Hamiltonian Monte Carlo (HMC) via the No-U-Turn Sampler (NUTS), which auto-adapts step size and path length, removing the need for manual tuning.

4

independent Markov chains

NUTS

via Numpyro backend (GPU-accelerated)

1,000+

warmup iterations (adaptation phase)

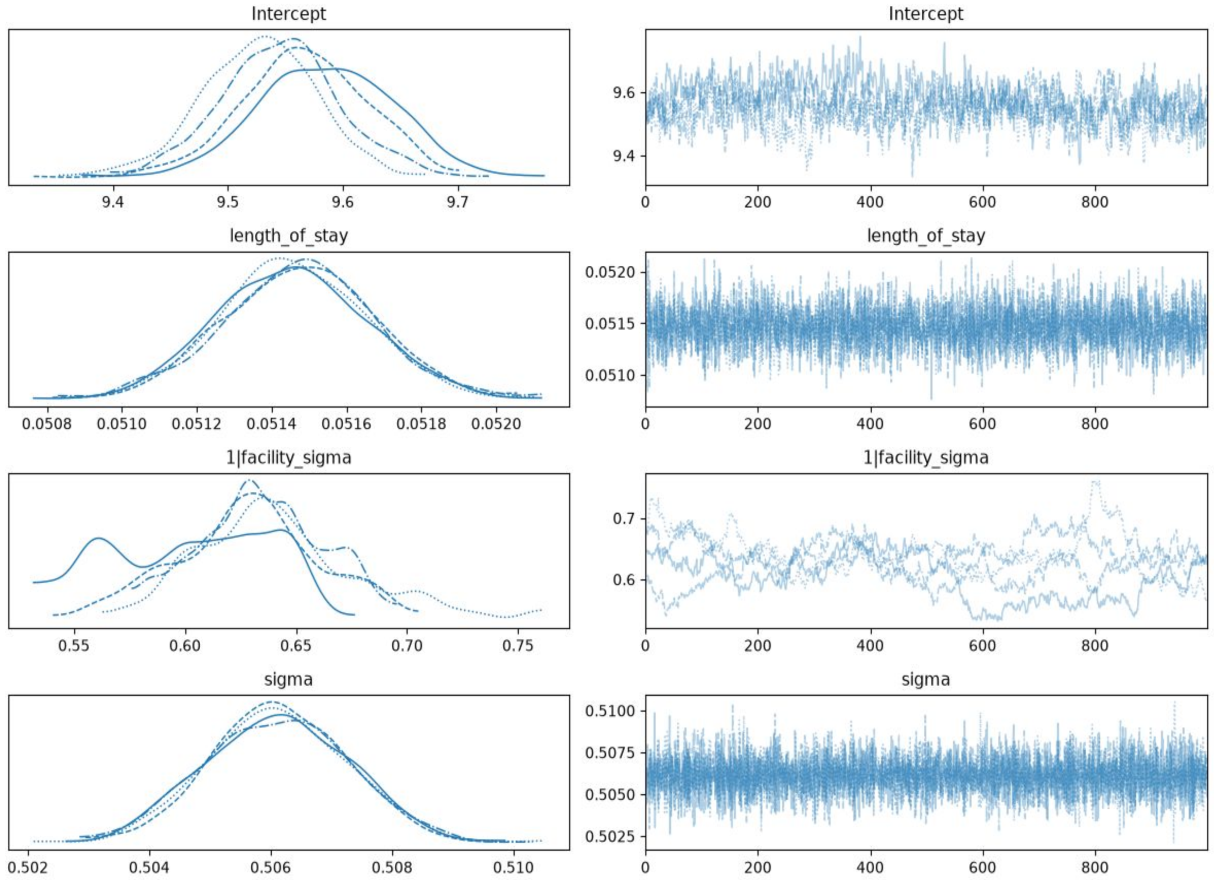
Sequential

chain execution — memory efficient

Why log-likelihood storage matters

Observation-level log-likelihood was stored for every fit (idata_kwargs={"log_likelihood": True}), enabling Bayesian model comparison via WAIC/LOO between the hierarchical and baseline models later in this deck.

The Convergence Journey



Trace plots from the initial (1,000-iteration) fit — chains for facility σ visibly fail to fully mix

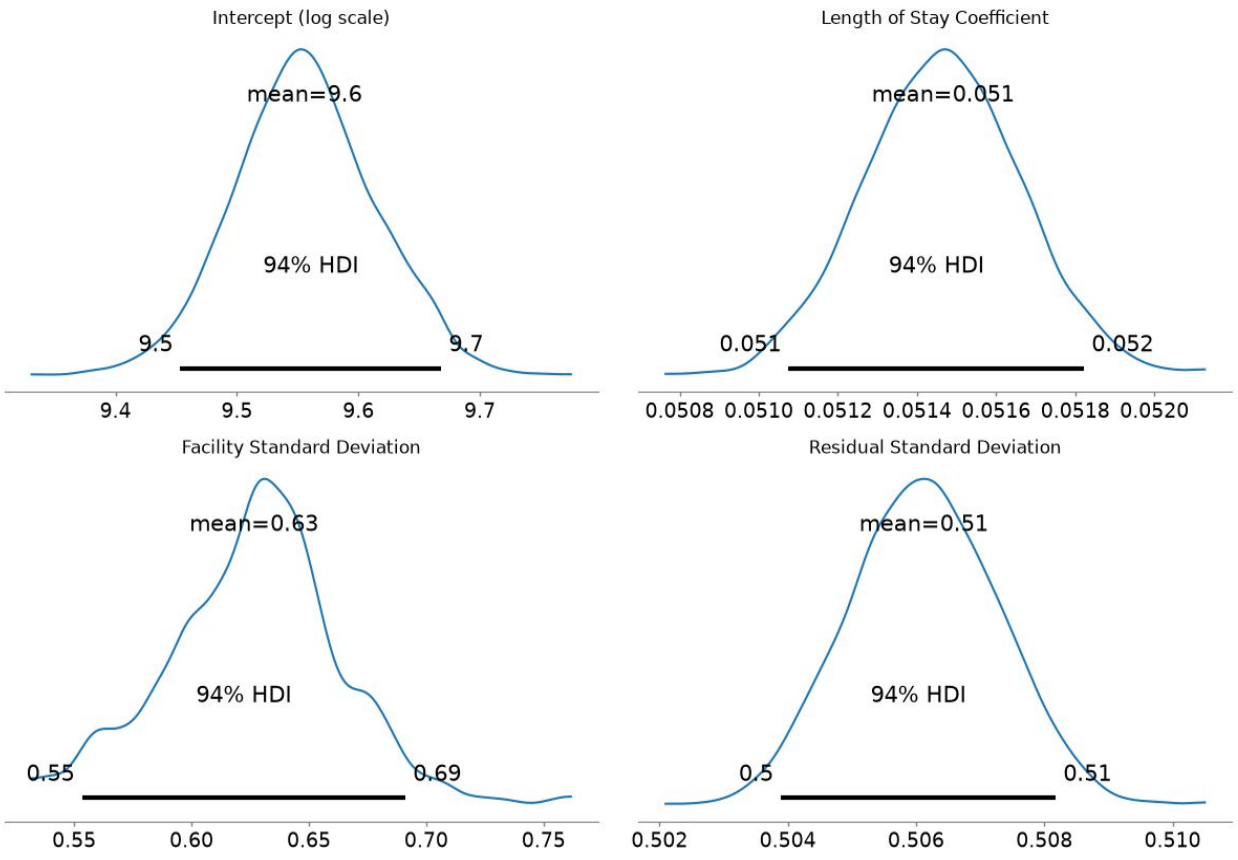
Parameter	1,000 iter	2,000 iter	4,000 iter
Intercept	1.15 ⚠️	1.03 ⚠️	1.04 ⚠️
length_of_stay	1.00 ✓	1.00 ✓	1.00 ✓
facility σ	1.27 ⚠️	1.03 ⚠️	1.03 ⚠️
σ (residual)	1.00 ✓	1.00 ✓	1.00 ✓

R-hat < 1.01 indicates full convergence — most parameters remained marginally above threshold across every iteration count tried.

Final decision: 2,000-iteration refit

Best convergence for the Intercept (R-hat 1.03, highest ESS at 107) with acceptable convergence overall (R-hat ≤ 1.03). 4,000 iterations showed diminishing returns at 2× the runtime — not worth the tradeoff.

Final Model: Posterior Estimates



Param	Mean	94% HDI	Meaning
`Intercept`	9.56	[9.45, 9.67]	Baseline log(charges)
`length_of_stay`	0.051	[0.051, 0.052]	+1 day → ~5.2% higher charges
`facility σ`	0.627	[0.553, 0.691]	Between-facility variation
`σ (residual)`	0.506	[0.504, 0.508]	Unexplained variation

Reading the facility σ

Facilities vary by roughly ±1.23 on the log scale after accounting for patient mix and length of stay — a substantial, genuine source of charge variation.

RESULTS

Facility-Level Shrinkage

Reading the plot

Each facility's estimated effect on log(charges), sorted low to high, with 94% uncertainty bars. The dashed red line marks the global mean.

5 / 201

facilities significantly different from average

2

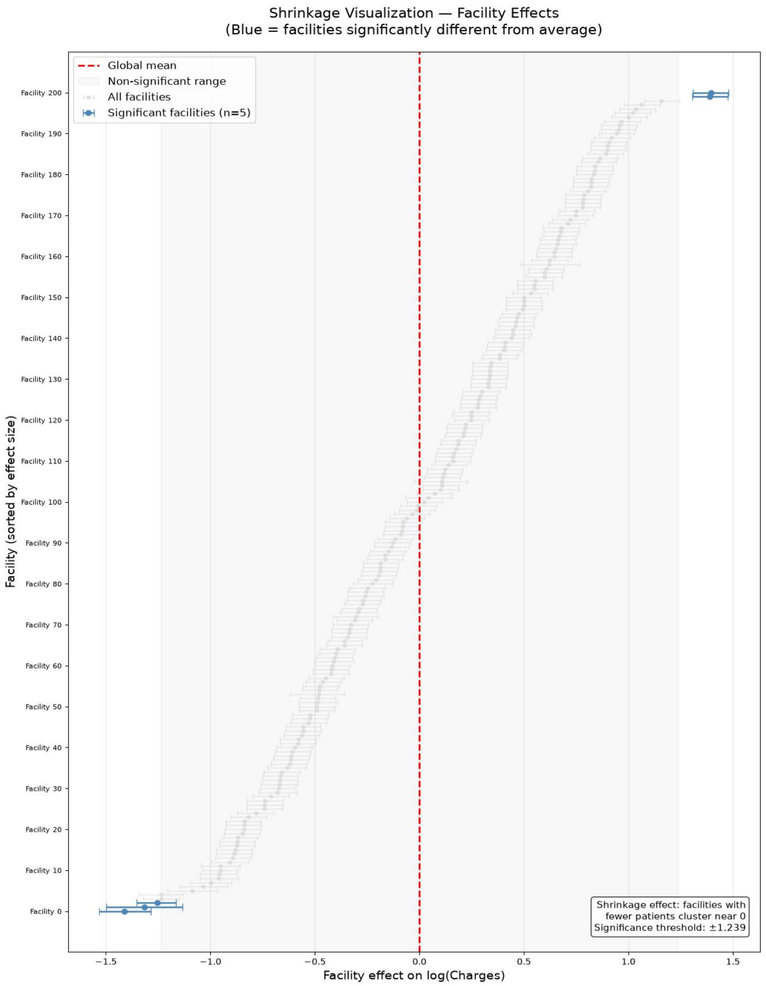
significantly higher-charging

3

significantly lower-charging

±1.239

significance threshold (log scale)



What this confirms

Most facilities cluster near the global mean with wide, overlapping uncertainty — exactly the shrinkage pattern partial pooling is designed to produce. A small handful stand out as genuinely, significantly different, even after adjusting for patient mix and length of stay.

Since length_of_stay is included as a predictor, this facility effect represents variation after accounting for how long each patient stayed — a cleaner read on facility pricing behavior itself.

Model Comparison: Hierarchical vs. Baseline

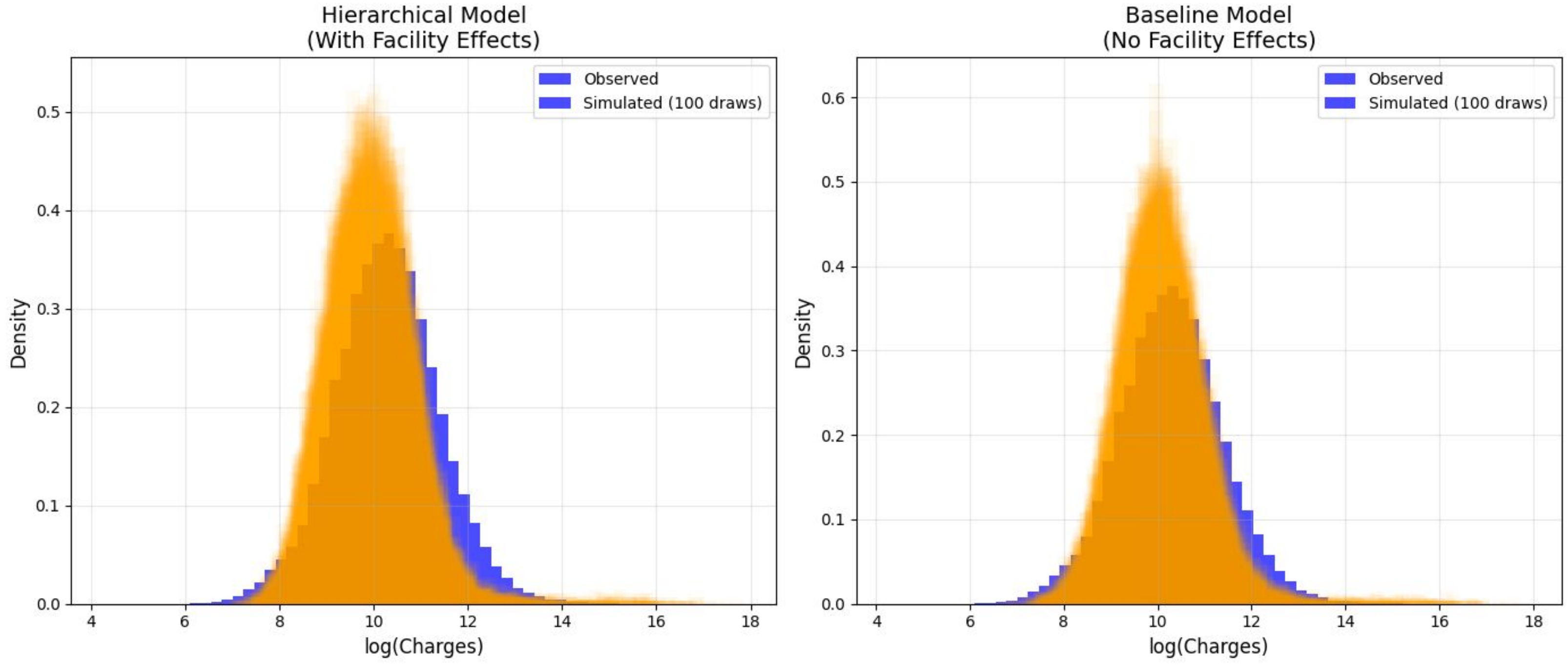
A baseline "flat" model — same fixed effects, no facility random intercept — was fit for comparison via WAIC (Watanabe-Akaike Information Criterion).

Model	elpd_waic	p_waic	Weight
Hierarchical	−69,695.5	228.0	0.978
Baseline (no facility)	−109,539.0	20.9	0.022

Decisive result

elpd_diff ≈ 39,844 — an enormous predictive-fit gap favoring the hierarchical model, backed by a 0.978 vs. 0.022 model weight split. Facility-level effects are not noise: they carry substantial, genuine predictive value beyond patient characteristics alone.

Posterior Predictive Checks



Simulated draws (orange) against observed log(charges) (blue) — both models track the observed shape closely

Metric	Hier.	Base.
Mean bias	0.340	0.194 ✓
Std-dev bias	0.179 ✓	0.195

The tradeoff

The baseline model predicts the mean slightly better; the hierarchical model captures the true spread in charges more accurately — because it distributes variance across facilities rather than absorbing it all as noise.

Given facility-level differences are central to our research question, the hierarchical model remains preferred.

LIMITATIONS

Where This Model Falls Short

Facility sampling cap

Large facilities were randomly capped at 500 records — this doesn't reflect true facility volume and could understate variance for high-volume hospitals.

Excluded predictors

APR MDC Description, Patient Disposition, and Risk of Mortality were dropped for parsimony or redundancy — some residual confounding may remain in the facility effect.

Imperfect convergence

R-hat stayed at or below 1.03 for the Intercept and facility σ across every iteration count tried, never fully reaching the < 1.01 target — estimates are stable but not textbook-clean.

Anonymized facility identity

Facility IDs carry no metadata (teaching vs. non-teaching, urban vs. rural, bed count) — we can measure that facilities differ, not fully explain why.

Homogeneous residual variance

A single σ is assumed across all facilities and severity levels; real charge variability likely differs by case complexity.

Association, not causation

The facility effect is best read as a pricing association, not a causal claim — unmeasured case-mix factors could still confound it.

KEY TAKEAWAYS

Facility Matters — and the Model Proves It

- Hierarchical model decisively outperforms a flat baseline (elpd_diff ≈ 39,844; weight 0.978 vs. 0.022)
- Facility charges vary by roughly ±1.23 on the log scale, even after adjusting for patient mix and length of stay
- 5 of 201 facilities are significantly different from the population average — the rest show the shrinkage pattern partial pooling is built for
- Each additional day of stay is associated with ~5.2% higher charges
- Convergence was imperfect (R-hat ≤ 1.03) but stable and consistent across every iteration count tested — a limitation worth noting, not a disqualifying one